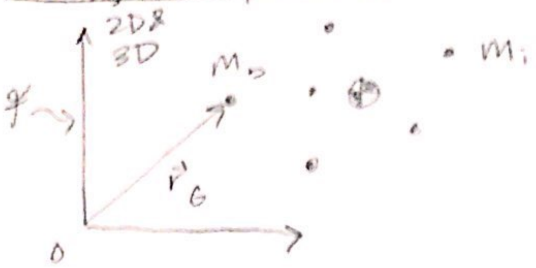
MAE 4730
9/19/18
Lecture 11

Today:
1) Energy theorems for systems of particles
2) DoF
3) Constraints

Energy of Systems:



CoM: $\boxed{m_{tot}\vec{r}_G = \sum m_i \vec{r}_i}$

E_k = Kinetic energy (extensive quantity)

$$\equiv \sum \tfrac{1}{2} m_i \underbrace{|\vec{v}_i|^2}_{\vec{v}_i \cdot \vec{v}_i}$$

Koenig's Theorem:

how to prove: $E_k = \dots \; \vec{v}_i = (\vec{v}_i - \vec{v}_G) + \vec{v}_G$
- multiply things out
- look for 0's

$$E_k = E_{kG} + E_{k/G}$$

$$= \underbrace{\tfrac{1}{2} m_{tot} V_G^2}_{\text{KE of CoM}} + \underbrace{\sum \tfrac{1}{2} m_i \underbrace{V_{i/G}^2}_{\vec{v}_{i/G} \cdot \vec{v}_{i/G}}}_{\text{KE wrt CoM}}$$

For energy, can't separate out internal + external forces

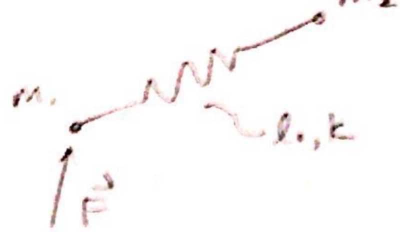
Work of $\underbrace{\text{all forces}}_{\text{int and ext.}} = \Delta E_k$

If there is a E_p:

$E_p(\vec{r}_1, \vec{r}_2, \dots)$

Work of nonconservative forces $= \Delta \underbrace{(E_p + E_k)}_{E_{tot}}$

ex]



$$\underbrace{\vec F_1 \cdot \vec v_1}_{\text{power of non conservative forces}} = \frac{d}{dt}\left[E_k + E_p\right] \quad \text{conservative forces}$$

E_p: $\frac{1}{2}k\left(|\vec r_1 - \vec r_2| - l_0\right)^2$

E_k: $\frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$

OR

$$\underbrace{\frac{d}{dt}\left(\frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2\right)}_{E_k} = \underbrace{\left(\vec F_1 + \frac{\vec r_2 - \vec r_1}{|\vec r_2 - \vec r_1|}\left(|\vec r_2 - \vec r_1| - l_0\right)k\right)\cdot \vec v_1 + \left(\frac{\vec r_1 - \vec r_2}{|\vec r_1 - \vec r_2|}\left(|\vec r_1 - \vec r_2| - l_0\right)k\right)\cdot \vec v_2}_{\text{Power of all forces}}$$

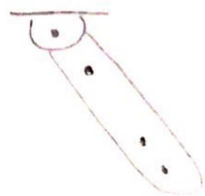
Koenig's Theorem: $\frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 = \frac{1}{2}(m_1 + m_2)\Big[\overbrace{\frac{m_1 \vec v_1 + m_2 \vec v_2}{m_1 + m_2}}^{\vec v_G}\Big]^2$
$+ \frac{1}{2}m_1|\vec v_1 - \vec v_G|^2 + \frac{1}{2}m_2|\vec v_2 - \vec v_G|^2$

Degrees of Freedom

DoF of a system

ex) Particle in $\begin{pmatrix} 1D \\ 2D \\ 3D \end{pmatrix}$ has $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ DoF

ex) Pendulum



has 1 DoF

ex) 1D

has 1 DoF

ex) 3D:

has

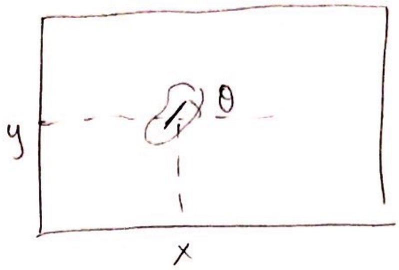
For a system # of DoF is the number of numbers needed to know
1) The position of every particle
2) The velocity given the particle

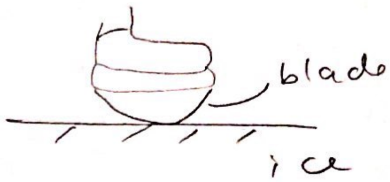
Non-holonomic mechanics is mech. of systems where

$vel_{DOF} < Pos_{DOF}$

ex) Look down on plane

figure skate





3 configurations available

2 velocity DoF

$3 > 2 \Rightarrow$ non holonomic · restricts velocity, not position

Constraints

ex) 1D

1DOF

ex) 3D

inextendable rod 5 DoF